Chapter B2: Keeping healthy

Overview

Issues of risk, ethics and social responsibility related to disease prevention and treatment in humans and plants are often in the news. Understanding the science of health and disease enables us to consider the issues critically, and to explore possible answers.

In Topic B2.1, learners explore how different pathogens are spread and cause disease, with reference to some common communicable diseases of humans and plants, then in Topic B2.2 they consider how the immune system in humans and plants protects against infection.

Topic B2.3 looks at ways in which individuals and society can reduce the spread of diseases, linked to issues of risk and decision making, for example with regard to vaccination.

Topic B2.4 develops understanding of ways in which diseases can be identified in the lab and in the field, and how new technologies offer the potential to improve lives.

In Topics B2.4 and B2.5, the way that lifestyle, environmental and genetic factors affect the risk of developing non-communicable diseases is explored, with reference to ideas about health studies, sampling, correlation and cause. Finally, learners learn about ways of treating diseases in Topics B2.5 and B2.6 and explore issues related to the development and testing of new medicines.

Learning about health and disease before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- appreciate that good hygiene helps humans keep healthy
- be able to identify and name the main parts of the human circulatory system, and describe the functions of the heart, blood vessels and blood
- appreciate the importance of bacteria in the human digestive system
- know that animals, including humans, need the right types and amount of nutrition, and that a healthy human diet includes carbohydrates, lipids (fats and oils), proteins, vitamins, minerals, dietary fibre and water
- recall some of the consequences of imbalances in the diet, including obesity, starvation and deficiency diseases
- recognise the impact of diet, exercise, drugs and lifestyle on the way their bodies function
- recall some of the effects of recreational drugs (including substance misuse) on behaviour, health and life processes.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about health and disease at GCSE (9–1)

B2.1 What are the causes of disease?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
The health of most organisms will be compromised by disease during their lifetime. Physical and mental health can be compromised by disease caused by infection by a pathogen, an organism's genes, environment or lifestyle, or trauma. Disease damages host cells and impairs functions, causing symptoms. However, an unhealthy organism may not always show symptoms of disease, particularly during the 'incubation period' after infection with a pathogen. Some diseases are communicable: they are caused by infection with pathogenic bacteria, viruses, protists and fungi, and can be spread from organism to organism in bodily fluids, on surfaces, and in food and water. Other diseases are non-communicable: they are not caused by infection but are associated with genetic, environmental and lifestyle factors. Some common diseases illustrate different types of pathogen and common routes of spread and infection, including: In humans: influenza (viral), *Salmonella* food poisoning (bacterial), Athlete's foot (fungal), malaria (protist) and HIV (viral STI). In plants: tobacco mosaic virus (viral), ash dieback (fungal) and crown gall disease (bacterial).	1. describe the relationship between health and disease 2. describe different types of diseases (including communicable and non-communicable diseases) 3. explain how communicable diseases (caused by viruses, bacteria, protists and fungi) are spread in animals and plants 4. describe common human infections including influenza (viral), Salmonella (bacterial), Athlete's foot (fungal) and malaria (protist) and sexually transmitted infections in humans including HIV/AIDS (viral) 5. describe plant diseases including tobacco mosaic virus (viral), ash dieback (fungal) and crown gall disease (bacterial)

Linked learning opportunities

Practical work:

- model the spread of infection using liquids (where one is 'infected' with an invisible chemical that can be detected experimentally)
- culture and microscopy of swabs from different surfaces

B2.2 How do organisms protect themselves against pathogens?

Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	
Humans have physical, chemical and bacterial defences that make it difficult for pathogens to enter the blood. These include the skin and mucus, stomach acid, saliva, tears, and bacteria in the gut. Platelets help to seal wounds to reduce the chance of pathogens entering the blood. These defences are always present, and are not produced in response to a specific pathogen. Plants have physical defences against pathogens, including the leaf cuticle and cell wall.	1.	describe non-specific defence systems of the human body against pathogens, including examples of physical, chemical and microbial defences
	2.	explain how platelets are adapted to their function in the blood
	3.	describe physical plant defences, including leaf cuticle and cell wall *(separate science only)*
The immune system of the human body works to protect us against disease caused by pathogens. White blood cells destroy pathogens. White blood cells have receptors that recognise antigens on pathogens, to distinguish between non-self and self. Different types of white blood cell are adapted to either ingest and digest pathogens, or produce antibodies to disable them or tag them for attack by other white blood cells. An antibody is specific for (only recognises) a particular antigen. Once the body has made antibodies against a pathogen, memory cells stay in the body to make antibodies quickly upon re-infection (immunity).	4.	explain the role of the immune system of the human body in defence against disease
	5.	explain how white blood cells are adapted to their functions in the blood, including what they do and how it helps protect against disease
Plants do not have circulating immune cells or produce antibodies, but they have a simple immune system that protects them against pathogens. For example, plants can make antimicrobial substances in response to pathogens. The ability of plants to protect themselves against pathogens is important in human food security.	6.	describe chemical plant defence responses, including antimicrobial substances *(separate science only)*

Linked learning opportunities

B2.3 How can we prevent the spread of infections?			
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*		***Linked learning opportunities***
Reducing and preventing the spread of communicable diseases in animals and plants helps prevent loss of life, destruction of habitats and loss of food sources. For plants, strategies include regulating the movement of plant material, sourcing healthy plants and seeds, destroying infected plants, polyculture, crop rotation, and chemical and biological control. For animals, including humans, strategies include vaccination (to establish immunity), contraception, hygiene, sanitation, sterilising wounds, restricting travel, and destruction of infected animals. The likely effectiveness, benefits, risks and cost of each strategy must be considered, and an individual's right to decide balanced with what is best for society (IaS4).	1.	explain how the spread of communicable diseases may be reduced or prevented in animals and plants, to include a minimum of one common human infection, one plant disease and sexually transmitted infections in humans including HIV/AIDS	**Practical work:** • investigate microbial growth on different foods and surfaces in different conditions. **Ideas about Science:** • discuss risk and decision making in the context of disease prevention (IaS4)
	2.	explain the use of vaccines in the prevention of disease, including the use of safe forms of pathogens and the need to vaccinate a large proportion of the population	

B2.4 How can we identify the cause of an infection? *(separate science only)*

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
In order to decide upon a course of treatment for a communicable disease, it is important to identify the disease and the pathogen causing it. There are standard ways to do this, including observing symptoms and taking samples of tissue or body fluid for cell counting, culture, microscopy, staining, testing with antimicrobials, and genome analysis. In addition, isolation and reinfection can be used to identify plant pathogens. Correct identification relies on use of aseptic techniques to avoid contamination of samples.	1. a) describe ways in which diseases, including plant diseases, can be detected and identified, in the lab and in the field b) describe how to use a light microscope to observe microorganisms *PAG1* 2. describe and explain the aseptic techniques used in culturing organisms *PAG7* 3. calculate cross-sectional areas of bacterial cultures and of clear zones around antibiotic discs on agar jelly using πr^2 M5c *PAG7*	**Practical work:** • investigate the effect of antibiotic discs on growth of microorganisms on agar plates • practice aseptic techniques
Monoclonal antibodies can be produced in the laboratory, using cultured clones of a white blood cell to produce antibodies against a particular antigen. All the antibodies produced by the clones recognise the same antigen. **New technologies using monoclonal antibodies are providing diagnostic tests (e.g. for diseases) with greater sensitivity and specificity. These tests give faster and more accurate results, which enables decisions (e.g. about treatment) to be made more quickly and based on more accurate information (IaS4).**	**4. describe how monoclonal antibodies are produced** **including the following steps:** • **antigen injected into an animal** • **antibody-producing cells taken from animal** • **cells producing the correct antibody selected then cultured** **5. describe some of the ways in which monoclonal antibodies can be used in diagnostic tests**	**Ideas about Science:** • **use of monoclonal antibodies as a technological application of science that could make a significant differences to people's lives (IaS4)**

B2.5 How can lifestyle, genes and the environment affect health?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Whether or not a person develops a non-communicable disease depends on many factors, including the genetic variants they inherited, their environment and aspects of their lifestyle. The interaction of genetic and lifestyle factors can increase or decrease the risk.	1. a) describe how the interaction of genetic and lifestyle factors can increase or decrease the risk of developing non-communicable human diseases, including cardiovascular diseases, many forms of cancer, some lung and liver diseases and diseases influenced by nutrition, including type 2 diabetes b) describe how to practically investigate the effect of exercise on pulse rate and recovery rate *PAG6*
	2. use given data to explain the incidence of non-communicable diseases at local, national and global levels with reference to lifestyle factors, including exercise, diet, alcohol and smoking
	3. in the context of data related to the causes, spread, effects and treatment of disease: a) translate information between graphical and numerical forms M4a b) construct and interpret frequency tables and diagrams, bar charts and histograms M4a, M4c c) understand the principles of sampling as applied to scientific data M2d d) use a scatter diagram to identify a correlation between two variables M2g
Different types of disease can interact, such as when having a disease increases or decreases the risk of developing or contracting another.	4. describe interactions between different types of disease

Linked learning opportunities

Specification links:

- what causes cancer (B4.3)
- diseases caused by genes (B1.3)

Practical work:

- investigate the amounts of fat and sugar in foods/drinks
- measure blood pressure, recovery rate

Ideas about Science:

- discuss correlation, cause and risk in the context of non-communicable diseases (IaS3, IaS4)

B2.6 How can we treat disease?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Humans have developed medicines that can control or eliminate the cause of some diseases and/or reduce the length or severity of symptoms. Antibiotics are becoming less effective due to the appearance of antibiotic-resistant bacteria. For non-communicable diseases such as cardiovascular diseases, strategies that lower the risk of developing the disease have benefits compared to treatments administered later. Many factors need to be considered when prescribing treatments, including the likely effectiveness, risk of adverse reactions and the costs and benefits to the patient and others (IaS4).	1. explain the use of medicines, including antibiotics, in the treatment of disease 2. calculate cross-sectional areas of bacterial cultures and of clear zones around antibiotic discs on agar jelly using πr^2 M5c *PAG7* 3. evaluate some different treatments for cardiovascular disease, including lifestyle changes, medicines and surgery	**Specification links:** • 'personalised medicine' (B1.3) • antibiotic resistance in microorganisms (B6.1) **Ideas about Science:** • risk and decision making in the context of medicines and treatment (IaS4)
Studying the genomes and proteins of pathogens and host cells can suggest targets for new medicines. Large libraries of substances are screened for their ability to affect a target. It is unlikely that a perfect medicine will be found during screening, but substances are selected for modification and further tests. All new medicines have to be tested before they are made widely available. Preclinical testing, for safety and effectiveness, uses cultured human cells and animals. Clinical testing uses healthy human volunteers to test for safety, and humans with the disease to test for safety and effectiveness. 'Open-label', 'blind' and 'double-blind' trials can be used. There are ethical questions around using placebos in tests on people with a disease (IaS4).	4. describe the process of discovery and development of potential new medicines including preclinical and clinical testing	**Ideas about Science:** • ethics in the context of using placebos in clinical testing of new medicines (IaS4)

B2.6 How can we treat disease?		
Some traditional treatments (e.g. radiotherapy and chemotherapy for cancer) cause adverse reactions. New technologies are enabling us to develop treatments that are more effective and have a lower risk of adverse reactions. For example, the specificity of monoclonal antibodies can be used to target cancer cells without damaging normal host cells.	**5.**	**describe how monoclonal antibodies can be used to treat cancer** **including:** • **produce monoclonal antibodies specific to a cancer cell antigen** • **inject the antibodies into the blood** • **the antibodies bind to cancer cells, tagging them for attack by white blood cells** • **the antibodies can also be attached to a radioactive or toxic substance to deliver it to cancer cells** *(separate science only)*

Ideas about Science:

- use of monoclonal antibodies as a technological application of science that could make a difference to people's lives (IaS4)